

PROGRAM 24:

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~
cd OSSP
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ mkdir -p Program24
cd Program24
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ pwd
/home/vedsk0831/OSSP/Program24
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ nano program24.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ gcc -Wall -Wextra -g program24.c -o program24
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ ls -l
total 24
-rwxr-xr-x 1 vedsk0831 vedsk0831 18416 Oct  6 14:27 program24
-rw-r--r-- 1 vedsk0831 vedsk0831  1841 Oct  6 14:27 program24.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ ./program24
=====
  OSSP PROGRAM 24 - FINAL DEMO
=====

Demo 1: Process execution
[START] Process Demo | PID=832
[DONE]  Process Demo | PID=832

Demo 2: Pipeline-style jobs
Pipeline Job 1 | PID=833
Pipeline Job 2 | PID=834
Pipeline jobs completed.

Demo 3: Signal handling
Signal demo process started | PID=835
Sending SIGTERM to child...
Signal demonstration completed.

=====
  FINAL PROJECT REVIEW COMPLETE
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ find . -maxdepth 2 -type f
./Makefile
./README.md
./program24.c
./program24
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ git status
fatal: not a git repository (or any of the parent directories): .git
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ git add
fatal: not a git repository (or any of the parent directories): .git
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program24$ cd ~/OSSP
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/vedsk0831/OSSP/.git/
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  Program16/
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Program16/
    Program18/
    Program19/
    Program20/
    Program21/
    Program22/
    Program23/
    Program24/

nothing added to commit but untracked files present (use "git add" to track)
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git add .
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
```

(use "git rm --cached <file>..." to unstage)

```
new file:   Program16/program16
new file:   Program16/program16.c
new file:   Program16/program17
new file:   Program16/program17.c
new file:   Program18/program18
new file:   Program18/program18.c
new file:   Program19/program19
new file:   Program19/program19.c
new file:   Program20/program20
new file:   Program20/program20.c
new file:   Program21/program21
new file:   Program21/program21.c
new file:   Program22/output.txt
new file:   Program22/program22
new file:   Program22/program22.c
new file:   Program22/test_program22.sh
new file:   Program22/valgrind.txt
new file:   Program23/program23
new file:   Program23/program23.c
new file:   Program24/Makefile
new file:   Program24/README.md
new file:   Program24/program24
new file:   Program24/program24.c
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git commit -m "Complete OSSP skill experiments"
Author identity unknown
```

*** Please tell me who you are.

Run

```
git config --global user.email "you@example.com"
```

*** Please tell me who you are.

Run

```
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
```

to set your account's default identity.

Omit --global to set the identity only in this repository.

fatal: empty ident name (for <vedsk0831@DESKTOP-83FU7HB.localdomain>) not allowed

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git tag -a v1.0 -m "OSSP Final Project v1.0"
```

fatal: Failed to resolve 'HEAD' as a valid ref.

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ git tag
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP$ |
```

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>

#include <signal.h>

void run_demo(const char *name, int seconds)
{
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return;
    }

    if (pid == 0) {

        printf("[START] %s | PID=%d\n",
            name, getpid());

        sleep(seconds);

        printf("[DONE] %s | PID=%d\n",
            name, getpid());

        exit(0);
    }

    waitpid(pid, NULL, 0);
}
```

```
int main()
{
    printf("=====\n");
    printf(" OSSP PROGRAM 24 - FINAL DEMO\n");
    printf("=====\n\n");

    printf("Demo 1: Process execution\n");
    run_demo("Process Demo", 1);

    printf("\nDemo 2: Pipeline-style jobs\n");

    pid_t p1 = fork();

    if (p1 == 0) {
        printf("Pipeline Job 1 | PID=%d\n", getpid());
        sleep(1);
        exit(0);
    }

    pid_t p2 = fork();

    if (p2 == 0) {
        printf("Pipeline Job 2 | PID=%d\n", getpid());
        sleep(2);
        exit(0);
    }

    waitpid(p1, NULL, 0);
```

```
waitpid(p2, NULL, 0);
```

```
printf("Pipeline jobs completed.\n");
```

```
printf("\nDemo 3: Signal handling\n");
```

```
pid_t child = fork();
```

```
if (child < 0) {  
    perror("fork");  
    return 1;  
}
```

```
if (child == 0) {
```

```
    printf("Signal demo process started | PID=%d\n",  
        getpid());
```

```
    sleep(5);
```

```
    printf("Signal demo completed.\n");
```

```
    exit(0);  
}
```

```
sleep(1);
```

```
printf("Sending SIGTERM to child...\n");
```

```
kill(child, SIGTERM);
```

```
waitpid(child, NULL, 0);
```

```
printf("Signal demonstration completed.\n");
```

```
printf("\n===== \n");
```

```
printf(" FINAL PROJECT REVIEW COMPLETE\n");
```

```
printf("===== \n");
```

```
return 0;
```

```
}
```